

Therefore, maximum number of intensity maxima observed is 5

20 A

The upward electric force F_E on each sphere must be equal and opposite to each of its weight. (i.e. $qE = mg$)

Since the same electric field E is experienced by both spheres, $\frac{q_x}{m_x} = \frac{q_y}{m_y} = \frac{g}{E}$

When the plates are brought closer together, E increases, F_E is now more than weight and the two charges will move upward.

The net force on each object is equal to the increase in F_E (i.e. $q\Delta E = ma$)

Since the charge to mass ratio $\frac{q}{m}$ of the two spheres is the same, the acceleration is also the same.